Sam (Peset) Tan

Senior Software Engineer

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SUMMARY

Senior software engineer building healthcare applications and shared platform infrastructure for medication-management workflows. Hands-on experience with Omnicell's OmniSphere platform, integrations between medication dispensing and inventory systems, restocking, medication-diversion workflows, and transactional auditing. Combines C#/.NET and distributed-systems development with technical leadership of three other engineers.

TECHNICAL SKILLS

Backend: C#, .NET, ASP.NET Core, TypeScript, Node.js, Python, Go

Systems and data: Kafka, PostgreSQL, ClickHouse, Redis, Elasticsearch, multitenancy, transactional outbox

Cloud and delivery: AWS, Kubernetes, Helm, Codefresh CI/CD, NiFi

Frontend: Angular, Nx, reusable components

Quality: Unit/integration/acceptance tests, concurrency testing, observability, performance benchmarking

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Omnicell | 2017-Present

Healthcare cloud software supporting medication workflows, dispensing integrations, and shared configuration, identity, and audit services.

- **Cloud modernization.** Served as the initial developer for a hospital restocking cloud application; helped migrate desktop workflows into three backend services and two Angular applications on Kubernetes. Owned scheduled-restocking design and much of its implementation, refining failure handling through manual and automated tests.
- **Shared audit infrastructure.** Implemented reusable ASP.NET Core auditing to record requests and changes to application data, keeping audit data isolated by customer and request and maintaining compatibility for services using the shared libraries. Improved audit middleware performance in a benchmark using an authenticated user with one role: 32.6% faster execution and 28.2% fewer memory allocations in repeated in-process .NET 10 Release-build tests.
- **Reliable event delivery.** Contributed to shared .NET/PostgreSQL/Kafka transactional-outbox infrastructure coordinating concurrent publishers, retrying failed deliveries, and preserving message order for each message key. Optimized database queries that select queued events for publishing; synthetic local tests across 100 tenants and three event-publishing workers reconciled 299,100 events with no observed loss, duplication, or ordering errors, while retaining at-least-once delivery semantics.

Omnicell | Senior Software Engineer (continued)

- **Audit ingestion and compatibility.** Implemented TypeScript/Node.js ingestion from tenant-specific Kafka topics into ClickHouse, validating incoming event data, generating consistent IDs for the same source event, and marking Kafka messages as processed only after saving their data. Retained legacy Elasticsearch integration and diagnosed cross-service delivery gaps; guided schema and message compatibility with existing systems.
- **Performance and production reliability.** Added performance monitoring for an edge synchronization service and changed blocking Kafka producer behavior to reduce message lag. Integrated Redis caching into shared configuration infrastructure and coordinated Elasticsearch reindexing with site reliability engineers through the production rollout to address capacity and operational-cost concerns.
- **Cross-service delivery.** Refined stock-area transfer designs and coordinated integration and testing across teams through initial release. Helped design and coordinate historical-data loading into newer systems for closed-loop inventory processing, spanning staging, enrichment, and progress tracking.
- **Full-stack engineering.** Built Angular/Nx audit-search features and reusable frontend components; preserved row selections when users changed pages and reset selection on filter or sort changes. Added role-change and login auditing, readable role names, and privacy-related metadata in events exchanged between services.
- **Technical leadership.** Technical lead for three other engineers since 2025 while continuing hands-on development. Review team-proposed designs, mentor engineers into broader backend ownership, and promote constructive feedback, learning from mistakes, and shared production support. Collaborate with product, architects, customers, and other teams on priorities and technical tradeoffs.

Web Developer and Data Analyst | ATAC | 2008-2017

- Developed C# tools to parse large aviation datasets, prepare Excel outputs, and integrate with proprietary Windows applications for analysts and airspace controllers.
- Cleaned and presented aviation data; prototyped Angular applications with Node.js backends to visualize flight, noise, weather, and historical trends.

EDUCATION

M.S. Software Engineering | Penn State

B.S. Electrical Engineering | San Jose State